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Paulina Rodriguez

Professional Summary

Ph.D. candidate in Mechanical and Aerospace Engineering specializing in verification, validation, and uncertainty quantification (VVUQ) for computational modeling and simulation. Experienced in discretization error analysis, model validation, and risk-informed credibility assessment for engineering applications. Proven ability to develop reproducible, HPC-enabled computational workflows for simulation and analysis, integrating tools such as OpenFOAM, ANSYS, and Python to support VVUQ and credibility assessment. Background includes regulatory and national laboratory experience supporting decision-focused modeling under uncertainty. U.S. Citizen with active DOE Q-level clearance.

Core Expertise

- VVUQ: Discretization error estimation (Richardson Extrapolation, Grid Convergence Index, StREEQ), validation under limited data (Area Validation Metric, modified area metric, tolerance intervals), uncertainty propagation (Monte Carlo, Latin Hypercube Sampling), sensitivity analysis (local and, global including Sobol' indices)
- Credibility & Risk-Informed Modeling: qualitative risk assessment, small sample model validation, regulatory-grade modeling using risk based gradations
- Computational Modeling & Simulation (CM&S): computational fluid dynamics, conjugate heat transfer, OpenFOAM,, ANSYS CFX, ALEGRA Multiphysics
- HPC & Reproducibility: SLURM, MPI, Snakemake orchestration, Git integrated testing, Docker containerization
- Programming: Python (NumPy, pandas, PyVista), Bash, Git

Education

Ph.D. Mechanical and Aerospace Engineering, The George Washington University (GWU), Washington, DC
Doctoral Candidate 2026

- Dissertation Proposal completed (05/03/2024)
  - Topic: *"Case Study: Enhancing Credibility and Reliability in Medical Device Evaluation through Computational Fluid Dynamics for Risk-Informed Decision-Making"*
- Doctoral Qualifying Examination completed (10/05/2022)
  - Topic: *"Machine Learning Scalability for Real World Data"*

M.S. Applied Mathematics, Claremont Graduate University (CGU), Claremont, CA 2012
B.A. Mathematics, University of California Santa Cruz (UCSC), Santa Cruz, CA 2010

Languages

English
Spanish

Technical Skills

Programming: Python, R, C++, C, Lua, PHP, JavaScript, Octave, YAML, JSON
Python Libraries: NumPy, SciPy, pandas, Matplotlib, PyVista, Snakemake
Scripting & Reproducibility: Bash, Git, Jupyter, LaTeX, Markdown, Docker Containers, Snakemake
High Performance Computing: MPI, OpenMP , Slurm
Simulation & Modeling: ANSYS CFX, OpenFOAM, CUBIT, ParaView, Alegra, PowerWorld-NAERM
Operating Systems: Linux, UNIX, macOS, Windows
Database Systems: PostgreSQL, MySQL

Professional Research Experience

Year-Round & Summer Intern: ***L-level security clearance*** **(Mentor: Brian Carnes, PhD)** **2022, 2023, 2024 – Current**
Sandia National Laboratories, 1544 VVUQ & Credibility Processes
Albuquerque, New Mexico

- **EFI 3D simulation verification:** Performed solution verification for multiphysics simulations using StREEQ, quantifying mesh convergence behavior and estimating numerical bias across operating conditions. Extended verification methodology from a 2D exploding-wire case to a 3D EFI simulation, enabling consistent discretization error assessment across model dimensionality and configurations. Conducted parameter and geometry sensitivity studies, evaluating the influence of resolution and model setup on solution accuracy and verification outcomes

- Exploding-wire 2D simulation verification: Performed solution verification for multiphysics simulations using StREEQ, quantifying mesh convergence behavior and estimating numerical bias across material models and operating conditions. Performed global parameter sensitivity studies using Sobol indices.
- Multi-metric validation for small sample datasets: Identified and implemented three methods: Area Metric, Modified Area Metric, and tolerance intervals with a python workflow.
- NAERM UQ workflow: Developed a scalable, reproducible UQ workflow for power grid resilience modeling using NAERM, integrating Snakemake, Python, and HPC resources to automate preprocessing, co-simulation setup, and post-processing. Implemented Latin Hypercube Sampling (LHS) to generate stochastic inputs for uncertainty propagation and automated co-simulation workflows, enabling structured evaluation of grid resilience under uncertainty. Computed and analyzed uncertainty in resilience metrics.
- Reproducibility: I developed and automated analysis using snakemake for python workflows and Next Generation Workflow SAW for automating HPC and VVUQ workflows.

Doctoral Candidate & DOE Computational Science Graduate Fellow (PI: Lorena Barba, PhD) — **2021 - Current**
The George Washington University (GWU) — Washington, DC

- Developed a **conjugate heat transfer (CHT) computational model** using OpenFOAM and ANSYS CFX for a medical device application
- Conducted **solution verification studies**, including mesh and timestep refinement, to quantify discretization error and observed order of accuracy
- Applied **Richardson Extrapolation, Grid Convergence Index (GCI), and statistical extrapolation (StREEQ)** to estimate numerical uncertainty and assess solution reliability
- Designed **validation studies under limited experimental data conditions**, applying a multimetric approach using tolerance intervals and area-based metrics.
- Built **reproducible HPC workflows** using Python, Snakemake, Containers, MPI, and SLURM for automated simulation and post-processing pipelines
- Integrated **risk-informed credibility assessment frameworks, ASME V&V40 Standard,** aligned with regulatory guidance to support decision-making
- Collaborated and managed FDA Subject Matter Experts (SME), employing agile project management to support communication, knowledge management, and subject matter expertise input.
- Published reproducible workflows and open-source tools to enhance transparency and robustness in computational science on Gitlab and figshare.
- Authored the proposal and secured the competitive FDA-NSF Scholar in Residence (SIR) & DOE funding.

Regulatory Science Research Scientist (Mentors: Tina Morrison & Matt R. Myers, PhDs) — **2017 - 2021**
U.S. Food and Drugs Administration (FDA) — Silver Spring, MD

- Led an Agile Approach to Risk-Informed Credibility (ABioM Project), applying risk-based methodologies to Computational Modeling & Simulation (CM&S) in regulatory contexts using commercial software ANSYS.
- Developed a **Python-based tool**, SimSight, to analyze large regulatory datasets (510(k), PMA), improving accessibility and efficiency of model assessment.
- Member of the *FDA Modeling & Simulation Working Group,* contributing to a comprehensive report on *Successes and Opportunities in Modeling & Simulation Practices*.
- Evaluated medical device product submission using CM&S assessing **predictive capability, uncertainty, and validation evidence** in support of regulatory decision-making.
- Trained by the International Association for the Engineering Modelling, Analysis & Simulation Community (NAFEMS) on VVUQ focusing on implementing these processes to ensure the credibility of CM&S.
- Provided Subject Matter Expertise to consult on regulatory reviews for medical devices, ensuring the credibility and reliability of computational models submitted for evaluation.
- Developed regulatory science research tools for medical device modeling while managing teams of 6 SMEs.
- Developed and implemented reproducibility, Python, and Git Training for FDA Interns.

Senior Web Developer **2015 - 2017**
Search Influence New Orleans, LA
- Maintained HTML, JavaScript, and Ruby on Rails code while improving data collection accuracy for 30 business accounts and search engine optimized more than 40 websites using advanced data analytics.

Graduate Researcher (PI: Claudia Rangel-Escareño, PhD) **2011**
National Institute of Genomic Medicine (INMEGEN, Spanish acronym) Mexico City, Mexico
- Used Bayesian statistical methods in R on a small sample large dataset of neurocysticercosis disease microarray dataset identifying overly expressed genes..

Graduate Researcher (Mentors: Ami Radunskaya, Ali Nadim, Claudia, Claudia Rangel-Escareño, PhDs) **2011 - 2012**
Claremont Graduate University (CGU) Claremont, CA
- Applied Bayesian statistical methods using R, to identify expressed genes in large allergen microarray data.
- Investigated the coffee ring effect analytically to characterize particle deposition.
- Designed an optimized solar chimney HVAC system in Matlab for an Environmental Design Group start-up.

Undergraduate Researcher (PI: Jorge Balbas, PhD; Mentors: Glenn E. Peterson, PhD) **2009**
Institute for Pure and Applied Mathematics, University of California, Los Angeles (UCLA) Los Angeles, CA
- Optimized orbit transfer using CM&S in C++ to improve mission efficiency for The Aerospace Corporation.

Undergraduate Researcher (PI: Debra K. Lewis, PhD) **2008 - 2009**
University of California, Santa Cruz (UCSC) Santa Cruz, CA
- Explored analytical methods for Hamiltonian systems.

Additional Work Experience

Tutor Coordinator **2014**
Learning Support Services, UCSC Santa Cruz, CA
- Managed tutor database and website, identified and ensured coverage of 158 courses requiring tutors. Recruited, hired, trained, mentored, and evaluated 250 tutors.

Program Assistant and Student Advisor **2013 - 2014, 2015**
California Teach Program, UCSC Santa Cruz, CA
- Managed program funding, organized events, advised student interns, modernized the program website, and streamlined the application processes electronically.

Contract Tutor **2013**
Youth Policy Institute Los Angeles, CA
- Provided one-on-one tutoring to 11 students (grades 1-7) in English Language Arts and Mathematics, achieving a 95% improvement rate (Los Angeles Unified School District).

Mathematics Department Tutor **2009**
Learning Support Services, UCSC Santa Cruz, CA
- Provided one-on-one tutoring for 8 undergraduate mathematics courses, supporting 10-15 students per 3 hour sessions. Achieving a 100% course pass rate among tutored students..

Co-Leader & Tutor **2007 - 2010**
Academic Excellence Program (ACE), UCSC Santa Cruz, CA
- Led tutoring sessions for upper division Mathematics courses to support underrepresented students.

Teaching & Volunteering

Judge, Central NM STEM Research Challenge in Energy and Transportation, UNM, Albuquerque, NM 2026
Invited Participant, Challenges in Next-Generation Ecosystems for Scientific Computing, Argonne National Lab 2025
Mentor, SIAM CSE25 Conference, Fort Worth, TX 2025
Instructor & Developer, Reproducibility, Python, and Git Training for FDA Interns at FDA, Silver Spring, MD 2023
Research Scientist, Annual STEM Day at Annapolis Middle School, Annapolis, MD 2022
Subject Matter Expert, FDA Digital Transformation OCR Search Capabilities, Silver Spring, MD 2020
Hill Day Participant, Association for Women in Mathematics, U.S. Capitol Washington, DC 2019
Subject Matter Expert, FDA CDRH's Experiential Learning Program, Dassault Systèmes, Waltham, MA 2019

Volunteer, USA Science & Engineering Festival: FDA Booth, Washington, DC 2018
Founder and Secretary, DC SACNAS Chapter, Washington, DC 2018
Volunteer, UNIDOS US: Hands-On Science Booth, Washington, DC 2018
Lead, Tech Talent South New Orleans: Kids Code New Orleans, New Orleans, LA 2016
Founder and President, SACNAS at the Claremont Colleges Chapter, Claremont, CA 2012
Council Member, Graduate Student Council, Claremont, CA 2011 - 2012
Tutor and Teaching Assistant, Harbor High School, Santa Cruz, CA 2010
Cal Teach Intern, California Teach (Cal Teach) Program at UCSC, Santa Cruz, CA 2008 - 2009
Mentor and Presenter, Expanding Your Horizons Conference, Santa Cruz, CA 2007 - 2008

Awards

Academic Program Annual, NNSA Office of Defense Programs, *Interview* 2023
DOE Computational Science Graduate Fellowship (CSGF) 2021 - 2025
FDA Outstanding Service Award 2019
Claremont Graduate University Math Tuition Fellowship 2008 - 2010
Koret UC LEADS Symposium Poster Recognition 2008 - 2010
CAMP Symposium Poster Recognition 2009 - 2010

Publications & Presentations

- **Rodriguez, P.**, Murray, B., Myers, M., Barba, L., "Credibility Assessment of Computational Modeling for Medical Device Applications: A Case Study: Focus on Validation", Publication to be Submitted 2026
- **Rodriguez, P.,** Carnes B., Myers, M., Barba, L., "Verification of a Conjugate Heat Transfer Model in OpenFOAM for a Simplified Medical Device Geometry", Publication to be submitted 2026.
- **Rodriguez, P.**, Dibaji, A, Myers, M., Morrison, T. "Agile for Biomedical Modeling (ABioM)", Publication to be re-submitted.
- Carnes, B., Damm, D., Ibanez-Granados, D., LLoyd J., **Rodriguez, P.,** "Validation of Exploding Wire Tests Using Alegra-NG", Sandia Technical Report, SAND2025-13767, Sandia National Laboratories, October 2025
- McInnes, L.C., Arnold, D., Balaprakash, P., Bernhardt, M., Cerny, B., Dubey, A., Giles, R., Hood, D.W., Leung, M.A., Lopez-Marrero, V., Messina, P., Newton, O.B., Oehmen, C., Wild, S.M., Willenbring, J., Woodley, L., Baylis, T., Bernholdt, D.E., Camano, C., Cohoon, J., Ferenbaugh, C., Fiore, S.M., Gesing, S., Gomez-Zara, D., Howison, J., Islam, T., Kepczynski, D., Lively, C., Menon, H., Messer, B., Ngom, M., Paliath, U., Papka, M.E., Qualters, I., Raybourn, E.M., Riley, K., **Rodriguez, P.**, Rouson, D., Schwalbe, M., Seal, S.K., Surer, O., Taylor, V., Wu, L. "Report of the 2025 Workshop on Next-Generation Ecosystems for Scientific Computing: Harnessing Community, Software, and AI for Cross-Disciplinary Team Science", Argonne National Laboratories Workshop Report, 2025, Internal Review.
- **Rodriguez, P.**, Position Paper, Challenges in Next-Generation Ecosystems for Scientific Computing Argonne Workshop, 2025
- **Rodriguez, P.**, Barba, L., "A Medical Device Case Study in Model Verification and Credibility", DOE CSGF Annual Program Review, 2025. Podium Presentation.
- **Rodriguez, P.**, Barba, L., "A Case Study on Assessing Credibility in Computational Models for Medical Devices. Focus on Solution Verification." SIAM CSE25, 2025. Podium Presentation.
- **Rodriguez, P.**, Barba, L., "Credibility Assessment of Computational Modeling for Medical Device Applications: A Case Study", NDiSTEM SACNAS 2024 Conference , 2024. Podium Presentation.
- **Rodriguez, P.**, Barba, L., "Medical Device Modeling: A Case Study in Building Credibility with Limited Reproducibility", DOE CSGF Annual Program Review, 2024. Poster.
- **Rodriguez, P.**, Barba, L., "Developing a Risk-Informed Computational Model for a Medical Device: A Credibility-Building Approach", DOE CSGF Annual Program Review, 2023. Poster.
- **Rodriguez, P.**, "Computational Models for Medical Devices: Building Trust", The GWU Mechanical and Aerospace Engineering Department Seminars, 2023. Podium Presentation.
- FDA. "Successes and Opportunities in Modeling and Simulation for FDA", Report on Modeling & Simulation at FDA, Report.
- **Rodriguez, P.**, Sarmakeeva, A., Barba, L., "Comparing Open-Source and Commercial Software Solvers for Hagen-Poiseuille Flow", DOE CSGF Annual Program Review, 2022. Poster.
- Sarmakeeva, A., **Rodriguez, P.**, Barba, L., "Verification of Open-Source and Commercial Numerical Solvers for Hagen-Poiseuille Flow", SEAS Student Research & Development Showcase, 2022. Poster.
- **Rodriguez, P.**, "Agile for Biomedical Modeling (ABioM)", FDA Presentation, 2019. Webinar.
- **Rodriguez, P.**, Dibaji, A, Murray, B., Myers, M., Pathmanathan, P., Morrison, T., "A Management Framework for Supporting Adaptive and Iterative VVUQ Efforts in Biomedical Modeling", ASME V&V Symposium, 2019. Podium Presentation.
- **Rodriguez, P.**, Dibaji, A, Murray, B., Myers, M., Morrison, T., "An Agile Verification and Validation Process for Generating Regulatory-Grade Evidence", ASME V&V Symposium, 2018. Podium Presentation.
- Fanger, M., **Rodriguez, P.**, Talacay, L., Takmakov, P., Morrison, T., "SimSight: Data Mining to Determine the Role of Computational Modeling & Simulation in Regulatory Decisions for Marketed Devices", 2019 FDA Summer Fellow Scientific Poster Day. Poster.

- Peterson, G.E., Campbell, E.T., Balbas, J., Ivy, S., Merkurjev, E., **Rodriguez, P.,** “Relative Performance of Lambert Solvers 1: 0-Revolution Methods, Adv Astronaut Sci”, 136 (1), pp. 1495-1510, presented at 20th AAS/AIAA Space Flight Mechanics Meeting, San Diego, CA, February 14-17, 2010. Conference Paper.
- **Rodriguez, P.**, Castaño, K., Rangel-Escareño, C., “High Feature-to-Sample Ratio Neurocysticercosis Data Set in Gene Expression Microarray Analysis”, 2011. Poster.
- **Rodriguez, P.**, Ivy, S., Merkujev, E., Hall, T., Balbas, J., “Implementing and Comparing Lambert Solvers for Trajectory Design Studies and Space Mission Analyses", Koret UC LEADS Symposium, CAMP Symposium, 2010. Poster.
- **Rodriguez, P.**, Villaron, M., Lewis, D., “Hamiltonian Systems and their Application to Dynamical Waves and Fluids”, NCUR, 2009, Koret UCLEADS Symposium, 2009, CAMP Symposium, 2009. Poster.